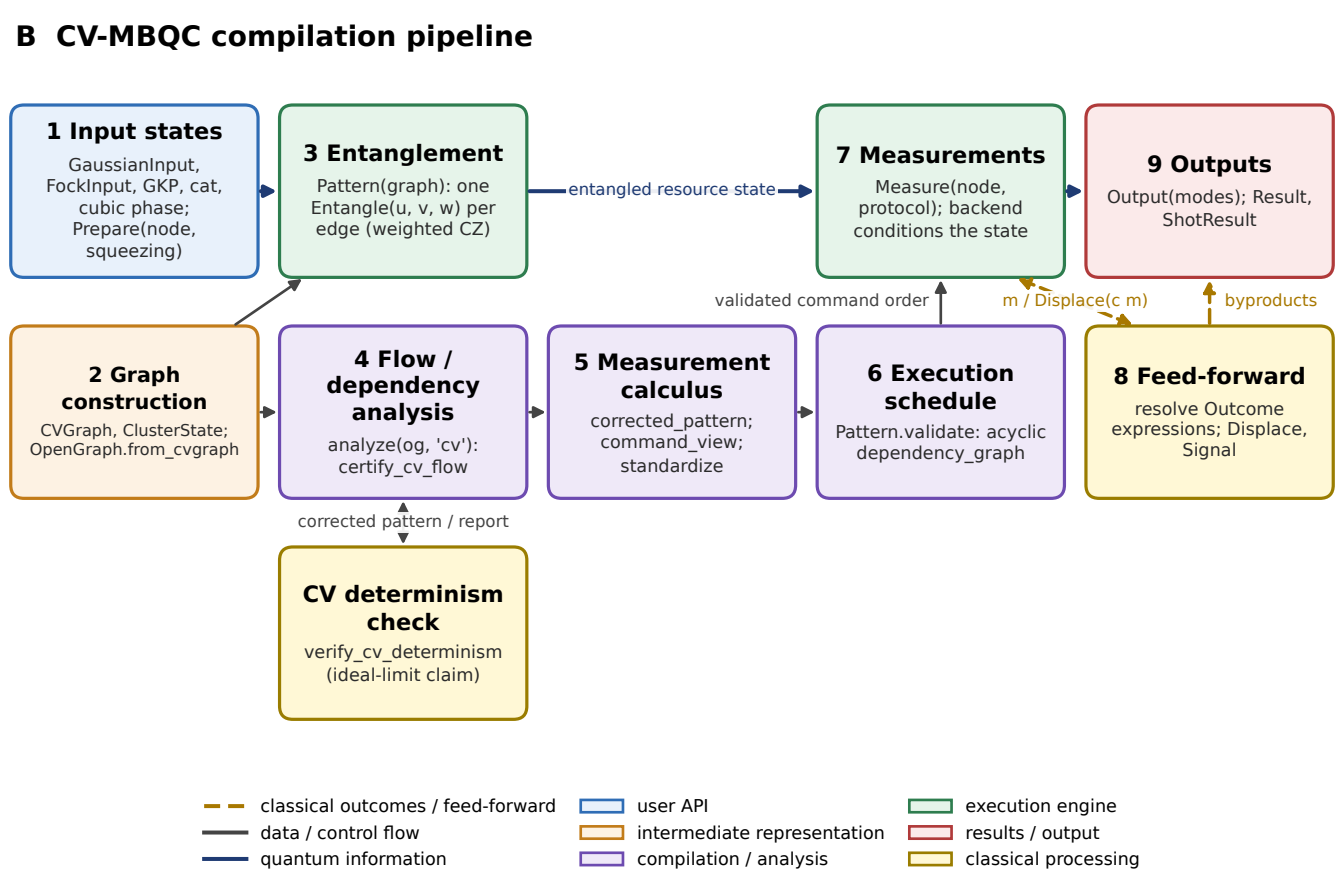
B CV-MBQC compilation pipeline
1 Input states
GaussianInput, FockInput, GKP, cat, cubic phase; Prepare(node, squeezing)
3 Entanglement
Pattern(graph): one Entangle(u, v, w) per edge (weighted CZ)
entangled resource state
7 Measurements
Measure(node, protocol); backend conditions the state
9 Outputs
Output(modes); Result, ShotResult
validated command order
m / Displace(c m)
byproducts
2 Graph construction
CVGraph, ClusterState; OpenGraph.from_cvgraph
4 Flow / dependency analysis
analyze(og, 'cv'): certify_cv_flow
5 Measurement calculus
corrected_pattern; command_view; standardize
6 Execution schedule
Pattern.validate: acyclic dependency_graph
8 Feed-forward
resolve Outcome expressions; Displace, Signal
corrected pattern / report
CV determinism check
verify_cv_determinism (ideal-limit claim)
classical outcomes / feed-forward
data / control flow
quantum information
user API
intermediate representation
compilation / analysis
execution engine
results / output
classical processing